1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights a convergence of immune suppression (PD-L1 upregulation on macrophages), EMT activation (SLUG/SNAI2 driven by paracrine PDGF■C), and growth signaling (mTOR) that could explain why organized immune infiltrates (e.g., TLS) do not always prevent recurrence/progression in bladder cancer. The puzzle is: under what conditions do TLS-like immune structures fail to provide durable control? 2) Tier 1 paths used: I built on the PD-L1 → macrophage → immune suppression axis (Core Path 1) and the PDGFC → SLUG EMT axis (Core Path 2), because these provide a direct mechanistic interaction between immune-extrinsic suppression and tumor-intrinsic plasticity that can blunt anti-tumor immunity. I did not put mTOR in the core hypothesis to keep the mechanistic question sharp, but it remains biologically relevant. 3) Tier 2 support: LZTS2's association with macrophage-linked processes and recurrence-free survival supports a prognostic link between myeloid biology and outcome; PDGFC reinforces the paracrine regulation of SLUG; confirmation that PD-L1 is myeloid-expressed strengthens the immune-suppression arm. These bolster the claim that macrophage PD-L1 and tumor SLUG together modulate TLS effectiveness. 4) Endpoint rationale: I selected PFI (progression-free interval) because it best maps to the Ontologist's recurrence-focused concern (recurrence-free survival) and is an objective clinical readout available in many cohorts for testing immune/EMT interactions. 5) Deliberate exclusions: I excluded mTOR from the core two-gene anchor to avoid diluting the mechanistic question (it becomes a broader signaling axis rather than the specific immune–EMT interaction). Other exploratory/ambiguous paths (surgical contraindications, AUR) were excluded as irrelevant to the mechanistic hypothesis about immune suppression and EMT-driven recurrence?

Working Hypothesis

In bladder cancer, TLS-like lymphoid aggregates predict longer progression-free intervals only in tumors that lack concurrent macrophage PD-L1 expression (CD274) and high SLUG (SNAI2) driven by PDGF-C signaling (PDGFC). Specifically, tumors with high TLS density but with high macrophage CD274 and high tumor SNAI2 will exhibit shorter PFI than TLS-high tumors without this dual immune-suppressive/EMT signature. The magnitude of this effect is modulated by PDGFC, MTOR, and LZTS2 signaling context, which influence EMT induction and macrophage phenotypes; thus, the interplay between TLS presence, macrophage PD-L1, and EMT determines the durable control of bladder tumors.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CD274, SNAI2
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

SNAI2: SNAI2 driven by paracrine PDGF■C), and growth signaling (mTOR) that could explain why organized immune infiltrates (e.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PDGFC	Supporting modifier from knowledge graph
Secondary	MTOR	Supporting modifier from knowledge graph
Secondary	LZTS2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective cohort analysis integrating TLS density, macrophage CD274 expression, SNAI2 levels, PDGFC, MTOR, LZTS2, and PFI;

Prediction 2: spatial and single-cell techniques (spatial transcriptomics, multiplex IHC/IF) to map co-localization of TLSs, PD-L1+ macrophages, and EMT markers;

Prediction 3: functional validation using in vitro co-culture models (bladder cancer cell lines with macrophages, PD-L1 and SLUG perturbations, PDGF-C stimulation) plus in vivo mouse models to assess TLS effects under PD-L1/EMT modulation. These approaches are feasible with current technologies and will illuminate whether TLS presence is conditionally predictive of PFI depending on the PD-L1/EMT axis.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PDGFC: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MTOR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.